

① Concept of Image Enhancement in Spatial domain:

Image enhancement is the process of improving the quality of an image so that it become easier for humans or computers to analyze. In the spatial domain, enhancement is performed by directly modifying the pixel value of the image.

Techniques:

1) Gray-Level Transformation:

- * changes the brightness and contrast of an image.

Application: medical images, x-ray images.

2) Histogram processing:

- * A histogram shows the distribution of pixel intensity values.

Application: satellite images, fingerprint recognition.

3) Spatial filtering:

- * smoothing filters to remove noise.

- * sharpening filters enhance edges and details.

Advantages:

- * Improve image quality

- * Enhance contrast and brightness

② D/B Image Smoothing and image sharpening:

Image smoothing and image sharpening are important image enhancement techniques. Smoothing removes noise, while sharpening improves image details.

Image Smoothing:

- * Reduces noise by averaging neighboring pixels.
- * produces a smoother image.
- * Uses mean filter, Gaussian filter, and median filter.

Applications:

- * medical imaging
- * digital photography
- * image preprocessing

Image Sharpening:

- * Enhances edges and fine details.
- * Uses Sobel, Prewitt and Laplacian filters.

Image Smoothing	Image Sharpening
Remove noise	Enhances edges
produces blur	produces sharp image
uses low-pass filter	uses high-pass filter
mean, Gaussian filter	Sobel, Laplacian filter

Image enhancement in the frequency domain:
Frequency domain enhancement modifies the frequency components of an image instead of directly changing pixel values. It uses Fourier Transform.

Low-pass filter (LPF)

- * Removes high-frequency noise.
- * produces smooth image
- * used for noise reduction.

Higher-pass filter (HPF)

- * Enhances high-frequency components like edges.
- * produces sharper image.
- * used for edge detection.

Comparison:

Low-pass filter	High-pass filter
Remove noise	Enhance edges
Smooth image	Sharp image
Remove high frequencies	Remove low frequencies

④ Image Segmentation :

Image Segmentation is the process of dividing an image into meaningful regions or objects for easier analysis.

Segmentation Techniques :

1. Point Detection :

- * Detects isolated pixels.
- * Used for fault detection.

2. Line Detection :

- * Detects horizontal, vertical and diagonal lines.
- * Used in road and document analysis.

3. Edge Detection :

- * Detects object boundaries.

4. Thresholding :

- * ~~Separates~~ Separates objects based on pixel intensity.
- * Simple and fast.

5. Region - Based Segmentation :

- * Groups neighboring pixels with similar properties.
- * Provides accurate segmentation.

Main the Importance of Edge Detection, Edge Linking and Boundary Detection.

Edge detection, edge linking and boundary detection are important image processing techniques used to identify objects and detect defects.

Edge detection :-

- * Detects sudden changes in pixel intensities.

Methods :- Sobel, Prewitt, Canny.

Edge linking :

- * Connects broken edges into continuous lines.
- * produces complete object shapes.

Boundary detection :

- * Finds the outer boundary of an object.
- * Helps measure object size and shape.

Applications:

- * Face recognition
- * Medical image analysis
- * Industrial defect detection.
- * Autonomous vehicle.
- * Traffic monitoring